

Coursework 2: Vector Space Semantics for Similarity between EastEnders Characters

Carlota Cortés Mir ; Student ID: 250711969

Question 1 (20 marks): Improve pre-processing

The baseline system used a very simple pre-process that only split text, giving a mean rank of about 4.31 on the validation set. To improve this, I added some pre-processing techniques to build a more complete preprocessing pipeline: lowercasing, tokenising with NLTK word tokenize, removing punctuation and numeric tokens, removing stopwords with the NLTK English list, and applying POS-aware lemmatisation, so inflected forms are collapsed to a shared lemma while keeping part-of-speech information. This first version reduced the mean rank down to about 1.9.

I then experimented with removing the `_EOL_` line markers and saw a further drop to 1.68, which suggests that `_EOL_` was acting more like noise than a useful structural feature. Next, instead of deleting all punctuation, I kept only a few expressive patterns and also selected some stopwords to not remove as they carry useful pragmatic information. With this final configuration, the mean rank improved again to about 1.62.

Question 2 (45 marks): Improve advanced feature extraction with Gen AI

For this question, I used a generative AI system (ChatGPT) to brainstorm feature-extraction ideas. I explained that each document contains all the lines from one EastEnders character, that similarity is computed with cosine over bag-of-words vectors, and asked which advanced features could improve performance at MSc level. I then followed up with more detail about my preprocessing and asked how different feature types would fit this pipeline. With that context, ChatGPT returned a ranked list of options and explained why each might help in this setup.

From these suggestions I chose four concrete techniques to implement:

1. TF-IDF weighting instead of raw counts
2. Word bigrams in addition to unigrams to capture short character-specific phrases
3. Character n-grams of length 4–5 extracted
4. POS-based aggregate features (ratios of nouns, verbs, pronouns, adjectives and modals) concatenated with the lexical vectors.

Each of these was tested both alone and in combination. The experiments showed that these more complex feature spaces did not consistently improve mean rank for this task. The best results still came from the simple unigram bag-of-words representation with strong preprocessing, even though Gen AI had recommended the more advanced combinations as a strong MSc-level answer.

The interaction highlighted a few things. Good prompts clearly helped because vague questions produced broad lists of techniques, while detailed prompts about the task, metric and existing pipeline led to more precise suggestions and reasoning. The model also always offered multiple options rather than a single “correct” solution, which was useful for exploring a menu of options and for justifying design choices, even when some features did not actually improve performance.

There were also some clear limitations. The model does not have direct access to the actual EastEnders data or the similarity matrix, so ideas that are normally reasonable (for example combining TF-IDF with character n-grams) did not always help in this cosine-similarity setup, which only became clear after testing on the held-out set. It also tended to suggest “advanced” feature combinations that made the vectors very sparse and weakened the frequent discourse words that are actually useful for recognising a character’s voice. Finally, Gen AI could not replace the practical work. All feature choices still had to be implemented, tuned and evaluated in Python by hand and integrated into the pipeline. The full ChatGPT history is included in a separate PDF, as required.

Question 3 (15 marks): Parameter Search

A systematic parameter search was carried out within Questions 1 and 2, with mean rank as the primary metric. For Q1, preprocessing options were varied incrementally: keeping vs removing EOL, different stopwords lists (standard vs selective), and with and without expressive punctuation tokens. The best-performing combination (lowercasing, NLTK tokenisation, punctuation/number removal, selective stopwords, expressive punctuation, POS-aware lemmatisation, no EOL) was fixed for later questions.

For Q2, a restricted grid search over feature extraction choices was performed using the fixed preprocessing. Again, each setting was evaluated on the validation set and compared via mean rank and accuracy. No cross-question grid search over all preprocessing and feature parameters was attempted, as this would have resulted in an unrealistic number of runs and was discouraged. Instead, optimal parameters were selected within each question and then fixed.

Question	Configuration	Mean rank
Q1	No preprocessing (split)	4.31
Q1	Lowercasing + NLTK tokenisation + punctuation/number removal + stopword removal + POS-aware lemmatisation	1.90
Q1	Above + remove <code>_EOL_</code> tokens	1.68
Q1	Above + selective punctuation & stopwords	1.625
Q2	Unigram counts	1.625
Q2	Unigrams + bigrams (counts)	2.00
Q2	Unigrams + TF-IDF	1.75
Q2	TF-IDF + bigrams + character n-grams + POS ratios	2.25

Table 1: Mean rank for different preprocessing and feature configurations.

Question 4 (10 marks): Analyse the similarity results

The similarity matrix between validation and training documents was inspected visually using the heatmap produced. For each validation character (row), the closest and furthest non-self training character was identified. This provided, for each character, an intuitive view of which other characters their held-out document most closely resembled. Generally, the closest non-self pairs correspond to characters who share storylines and speaking style, so their lemmatised word distributions and remaining function words look very similar. For example, pairs such as IAN-JANE, JACK-MAX, SEAN-MAX and TANYA-MAX appear as bright off-diagonal cells, which fits their overlapping family or work-related plots and similar everyday vocabulary. In contrast, the darkest off-diagonal cells identify pairs that are furthest apart, such as JANE vs SHIRLEY or PHIL vs OTHER, where characters differ more in topic and tone, leading to less overlap in lemmas and expressive markers.

Question 5 (10 marks): Run on final test data

Finally, the best system identified in the earlier questions was trained on the training data and evaluated on the held out test set. On the test set, the system achieved:

- Mean rank: 2.5
- Mean cosine similarity: 0.7944
- Accuracy: 8 correct out of 16 characters (0.5)

On the test set the system is weaker than on validation, the correct character is ranked 1st for only 8/16 cases (accuracy 0.5) and the mean rank increases from 1.62 on validation to 2.5 on test. At the same time, the mean cosine similarity between each test character and its own training vector remains relatively high (0.79), which suggests that test and training documents for the same character are still close in the vector space, but in some cases another character is slightly closer.